CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in agentic AI systems with orchestration, tool use and human-in-the-loop review, from multi-agent LLM pipelines and knowledge graphs to production deployment on Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector at an early-stage start-up, backed by a strong foundation in machine learning on sensor time-series and Python development.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an agentic GenAI knowledge-extraction system end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and implemented a Model Context Protocol (MCP) query service that connects the knowledge graph to AI agents with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Designed and optimized multi-agent LLM workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, with evaluation against an SAP-system-derived ground truth raising precision/recall from 33%/45% to 93%/90%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts review and approve knowledge-graph changes before they are applied.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, agentic AI, MCP, LangChain, LiteLLM, FastAPI, Postgres, Neo4j (knowledge graph), React, Azure, Kubernetes, Docker, MongoDB, RAG, GraphRAG**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to map their decision-making workflows and translated the requirements into an agentic AI application with LangGraph, deployed to AWS, that routes each request across vector database queries, general LLM inference and summarization.
- Developed a Python backend with FastAPI REST APIs and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK (infrastructure as code), following clean code practices; versioning in GitLab with CI/CD deployment.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated LLM evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Plant Maintenance Agent | LangGraph | FastAPI | PostgreSQL | OpenCV | Isolation Forest

- Multi-agent predictive maintenance assistant on simulated pump sensor data (temperature, vibration with FFT features, OPC UA-style tags): an anomaly agent, a manual-search agent and an orchestrator that proposes maintenance actions, with an approval gate for a human reviewer, tracing and per-run cost logging.
- A multimodal computer vision check with OpenCV and a pretrained detector on public surface-defect images, FastAPI service with PostgreSQL work-order storage, in Docker, evaluation set run in CI.

TECHNICAL SKILLSET

- **Agentic AI & GenAI:** agentic AI, LangGraph (multi-agent orchestration), tool use, human-in-the-loop, MCP, LangChain, LiteLLM, RAG, GraphRAG, LLM evaluation (RAGAS, PytrecEval), QLoRA fine-tuning
- **Backend & APIs:** Python, FastAPI, PostgreSQL, REST APIs, Python microservices, Postman, React (beginner)
- **Cloud & Operations:** Azure, Azure AI Search, AWS (CDK, Lambda, SQS, DynamoDB, EC2), Docker, Kubernetes, Git, GitLab CI/CD
- **Data & ML:** time-series and sensor data (XGBoost, LSTM, Scikit-Learn), PyTorch, Pandas, Neo4j (knowledge graph), SQL, MongoDB, Vector Databases (Qdrant, Pinecone)
- **Ways of working:** AI-assisted development (Claude Code), clean code, Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI